

Metabolomics Biomarker Analysis Report

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1 Executive Summary

This report presents the results of machine learning analysis for metabolomics biomarker discovery.

1.1 Key Findings

Table 1: Analysis Summary

Metric	Value
Best Model	Ensemble
Test Accuracy	0.619
Test Kappa	0.236
Number of Features	5
Best Threshold	0.3

2 Model Performance

2.1 Model Comparison

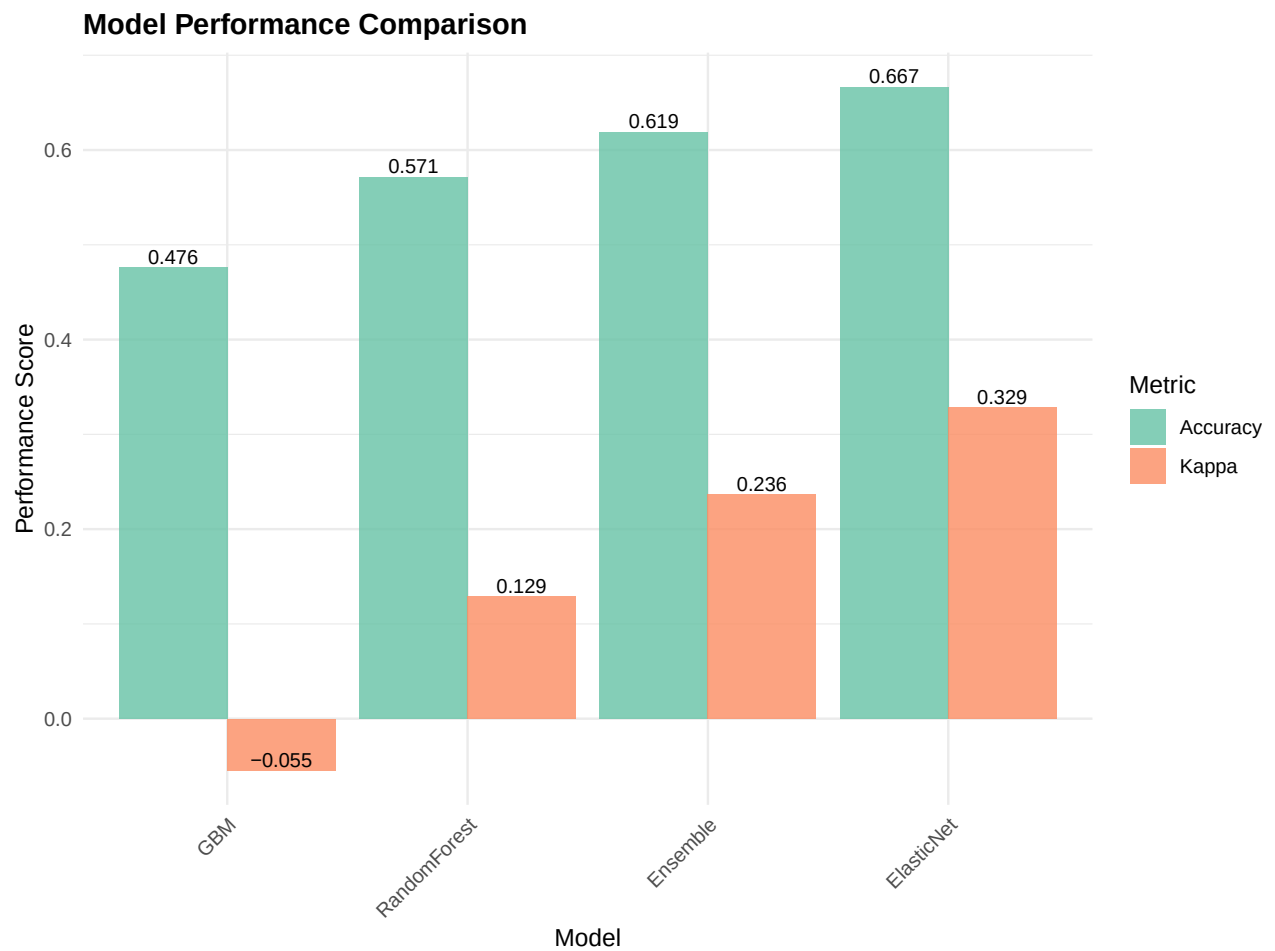


Figure 1: Performance comparison across different machine learning models

Table 2: Detailed Model Performance Metrics

	Model	Accuracy	Kappa	Best_Threshold
ElasticNet.Accuracy	ElasticNet	0.667	0.329	0.3
Ensemble.Accuracy	Ensemble	0.619	0.236	0.3
RandomForest.Accuracy	RandomForest	0.571	0.129	0.3
GBM.Accuracy	GBM	0.476	-0.055	0.3

3 Biomarker Discovery

3.1 Top Metabolite Biomarkers

Table 3: Top 10 Metabolite Biomarkers by Importance

Rank	Metabolite	Relative Importance (%)
1	Haematommic acid	36.62
2	Glycerol Tributyrat	31.19
3	Mannitol	13.41
4	Ribonolactone	12.03
5	OLEOYL GLYCEROL	6.74

3.2 Partial Dependence Analysis

The following plots show how each metabolite concentration affects the predicted probability of the outcome.

4 Model Evaluation

4.1 Prediction Distribution

4.2 Confusion Matrix

Table 4: Classification Performance Metrics

	Metric	Value
Accuracy	Accuracy	0.619
Kappa	Kappa	0.236
Sensitivity	Sensitivity	0.636
Specificity	Specificity	0.600
Pos Pred Value	Pos Pred Value	0.636
Neg Pred Value	Neg Pred Value	0.600

5 Feature Relationships

5.1 Metabolite Correlations

6 Conclusions

The ensemble model achieved **61.9%** accuracy with a Kappa score of **0.236**, indicating fair agreement beyond chance.

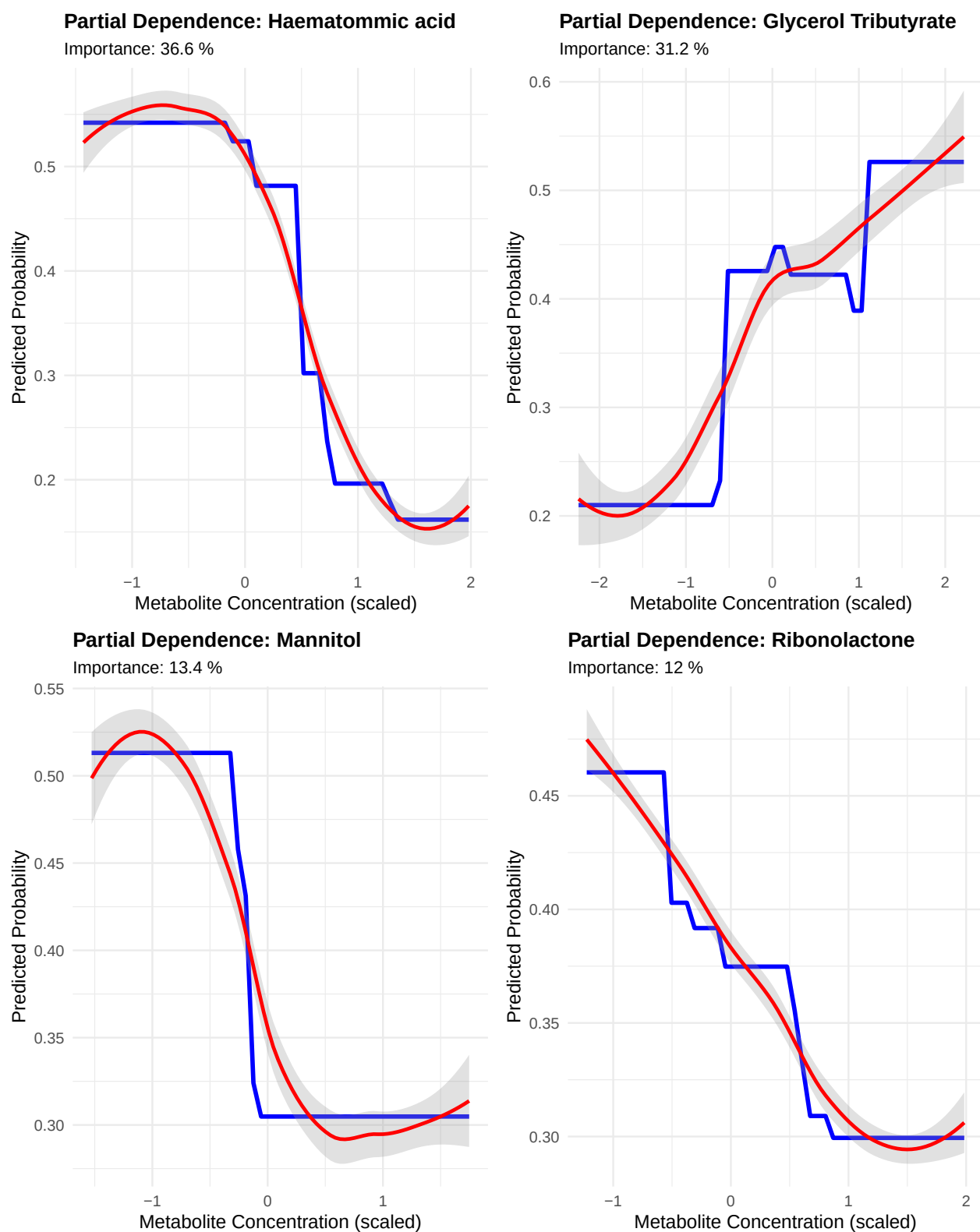


Figure 2: Partial dependence plots for top 4 metabolites

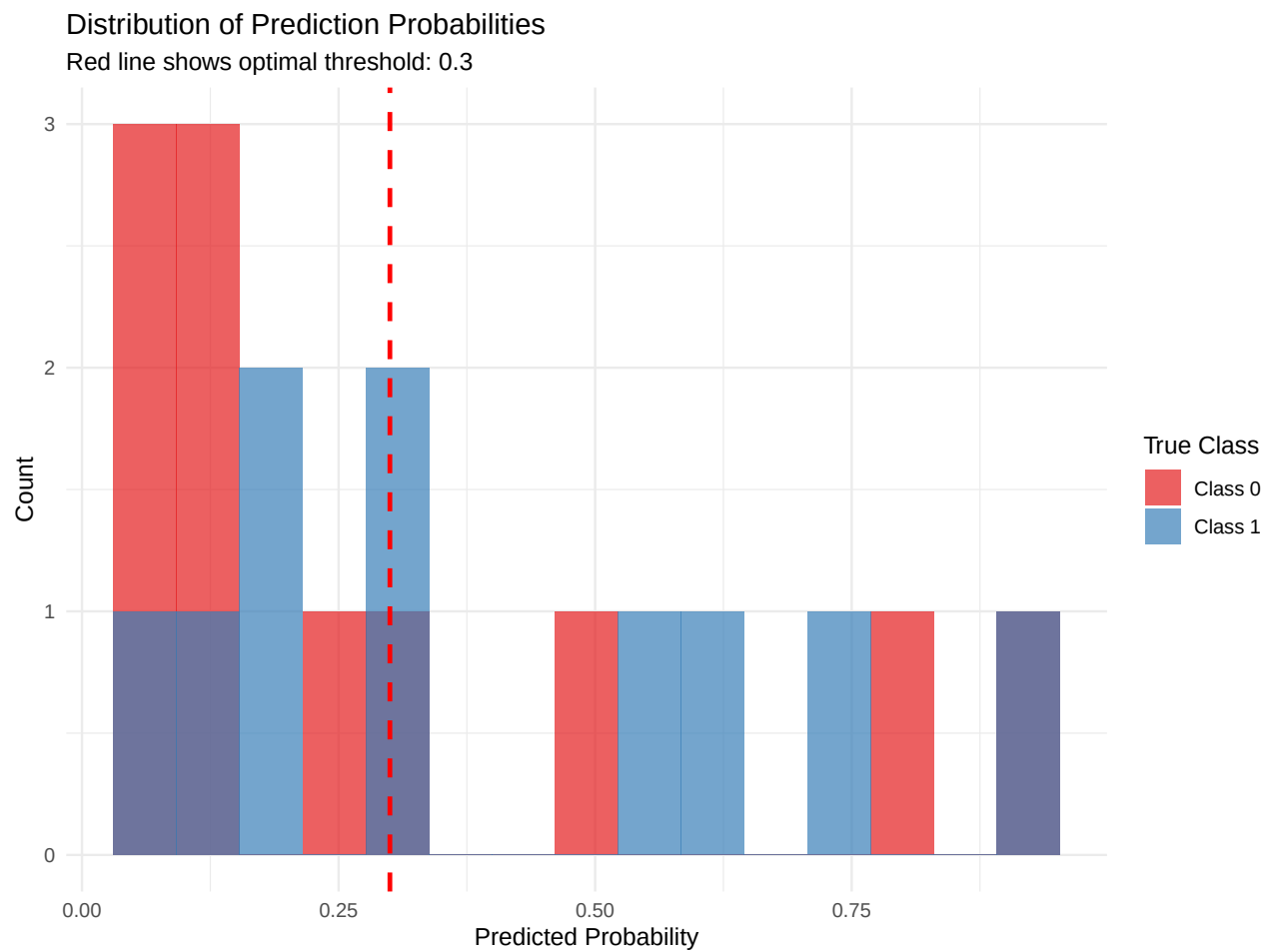


Figure 3: Distribution of prediction probabilities by true class

Feature Correlation Matrix

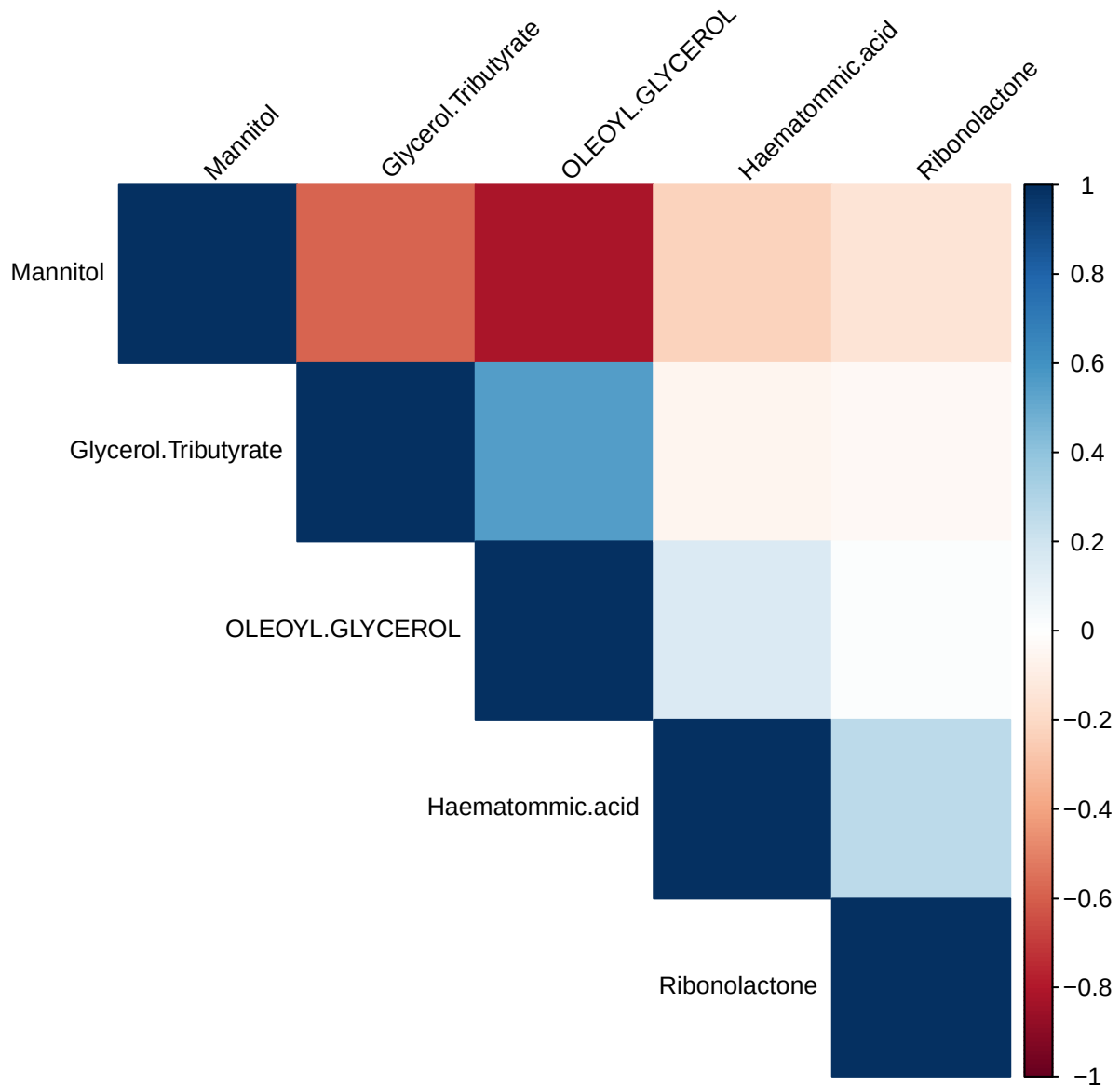


Figure 4: Correlation matrix of selected metabolites

The top biomarker metabolites identified were: - Haematommic acid - Glycerol Tributyrate - Mannitol - Ribonolactone - OLEOYL GLYCEROL

These results suggest promising metabolite biomarkers for further validation studies.